

BILINEAR RADIAL SCALING OVER FINITE FIELDS

ANONYMOUS

ABSTRACT. Let B be a nondegenerate bilinear form on $V = \mathbb{F}_q^m$. We determine the finite polynomial map $\Phi(u, v) = (B(u, v)u, B(u, v)v)$. Its t th iterate is radial with exponent $(3^t - 1)/2$, while its bilinear value follows the scalar cube map. This gives every tail and eventual period from the 3-primary part of the initial value. We lift the scalar description back to all pair targets: every-time fibres are power-map root counts, except for a null cone collapsing to zero. We also give every tail population, the recurrent census, image size, and complete one-step fibre distribution; zero is the unique maximum-fibre target. Power maps, bilinear level counts, and generic polynomial functional graphs receive zero contribution credit. Status is OWNER_AMBER / HOLD_EXTERNAL.

1. LITERAL MAP AND SUBTRACTION BOUNDARY

Fix a nondegenerate bilinear form $B : V \times V \rightarrow \mathbb{F}_q$ and define

$$(1) \quad \Phi(u, v) = (cu, cv), \quad c = B(u, v).$$

The form need not be symmetric. A state means the ordered pair (u, v) ; write $0 = (0, 0)$.

Monomial dynamics over finite fields is established background [1], as are polynomial functional graphs [2] and, most directly, the structure of power-map functional graphs over finite groups [3]. We therefore assign finite-field cyclicity, scalar power-map tails and roots, and static level counts of B zero contribution credit. The retained conjunction is the literal pair map (1), its radial iterate, the lift of every scalar time/fibre statement to every pair target, the exceptional null cone, and the complete population atlas. A literal or conjugate owner triggers withdrawal. The bounded source non-hit used for internal drafting is not a novelty certificate.

For later counts put

$$(2) \quad Q = q^{m-1}(q^m - 1), \quad Z = q^{2m-1} + q^m - q^{m-1}.$$

Lemma 1.1 (bilinear levels). *For each $d \in \mathbb{F}_q^*$, exactly Q states satisfy $B(u, v) = d$, while exactly Z states lie on the null cone $B(u, v) = 0$.*

Proof. For each $u \neq 0$, nondegeneracy makes $v \mapsto B(u, v)$ a nonzero linear functional, so every value has q^{m-1} preimages. Thus a nonzero level has $(q^m - 1)q^{m-1} = Q$ points. Subtracting the $q - 1$ nonzero levels from q^{2m} gives Z . $\square$

2. ITERATES, TAILS, AND PERIODS

Set $a_t = (3^t - 1)/2$.

Theorem 2.1 (closed iterate and scalar factor). *For every $t \geq 0$,*

$$(3) \quad \Phi^t(u, v) = (c^{a_t}u, c^{a_t}v), \quad B(\Phi^t(u, v)) = c^{3^t},$$

with the evident exponent-zero convention at $t = 0$.

Proof. If a pair is scaled by λ , bilinearity scales its B -value by λ^2 . Hence one step sends c to c^3 . If the accumulated radial exponent is a_t , the next is $a_t + 3^t = a_{t+1}$. Induction proves both identities, including $c = 0$. $\square$

For $c \neq 0$, let $r = \text{ord}_{\mathbb{F}_q^*}(c)$ and factor $r = 3^a s$ with $(s, 3) = 1$. Write $\text{ord}_{2s}(3)$ for the multiplicative order of 3 modulo $2s$; in particular $\text{ord}_2(3) = 1$.

Theorem 2.2 (pointwise functional graph). *The zero state is fixed. Every other state on the null cone has exact tail one and eventual period one. A state with $c \neq 0$ has exact tail a and eventual period*

$$(4) \quad \text{ord}_{2s}(3).$$

Proof. A null state maps to zero, proving the first claims. For $c \neq 0$, the pair itself has a nonzero coordinate, so two radial multiples are equal exactly when their scalars are equal. By (3), equality at epochs t and $t + \ell$ is equivalent to

$$(5) \quad 3^a s \mid 3^t \frac{3^\ell - 1}{2}.$$

Because $3 \nmid 3^\ell - 1$, a cycle can first be reached exactly at $t = a$. After cancelling 3^a , the least positive ℓ satisfies $2s \mid 3^\ell - 1$, which is (4). $\square$

3. EVERY-TIME FIBRES

Theorem 3.1 (all targets at all positive times). *Fix $t \geq 1$, put $g_t = \gcd(3^t, q - 1)$, and let (x, y) be a target with $d = B(x, y)$. Then*

$$(6) \quad \#(\Phi^t)^{-1}(x, y) = \begin{cases} Z, & (x, y) = 0, \\ 0, & (x, y) \neq 0 \text{ and } d = 0, \\ g_t, & d \neq 0 \text{ and } d^{(q-1)/g_t} = 1, \\ 0, & d \neq 0 \text{ and } d^{(q-1)/g_t} \neq 1. \end{cases}$$

Proof. The null cone is precisely the fibre of zero: its points collapse in one step, while a nonzero bilinear value gives a nonzero radial scalar. Now let the target be nonzero. A predecessor with initial value $c \neq 0$ is forced by (3) to be $c^{-a_t}(x, y)$. Its bilinear value equals c exactly when $d = c^{2a_t+1} = c^{3^t}$. Thus predecessors are in bijection with 3^t th roots of d in the cyclic group $\mathbb{F}_q^*$. The standard cyclic-group root criterion gives the final two cases. $\square$

This proof supplies an explicit predecessor, not just its number: each root c gives $c^{-a_t}(x, y)$.

4. POPULATION AND IMAGE ATLASES

Write $q - 1 = 3^A h$ with $(h, 3) = 1$.

Theorem 4.1 (complete tail census). *Among states with nonzero B-value, exactly hQ are recurrent, and for $1 \leq a \leq A$ exactly*

$$(7) \quad 2 \cdot 3^{a-1} hQ$$

have tail a . Including the null cone, the total tail-zero population is $1 + hQ$; the tail-one population is $Z - 1 + \mathbf{1}_{A \geq 1} 2hQ$; and for $2 \leq a \leq A$ it is (7). The sharp maximum tail is $\max\{1, A\}$.

Proof. The cyclic group $\mathbb{F}_q^*$ is the product of cyclic factors of orders 3^A and h . Exactly h elements have order prime to 3. For $a \geq 1$, exactly $\varphi(3^a)h = 2 \cdot 3^{a-1}h$ elements have order with exact 3-primary part 3^a . Multiply these scalar counts by the common level size Q from Lemma 1.1, then add zero and the $Z - 1$ nonzero null states. Such null states ensure tail one exists for every q, m . $\square$

Theorem 4.2 (image and one-step fibres). *Let $g = \gcd(3, q - 1)$. Then*

$$(8) \quad |\text{im } \Phi| = 1 + \frac{(q-1)Q}{g}.$$

Every nonzero image target has exactly g predecessors. Zero has Z predecessors and is the unique maximum-fibre target.

Proof. Apply Theorem 3.1 at $t = 1$. The cube subgroup has $(q-1)/g$ values, each supporting Q pair targets, plus zero. This proves (8) and uniformity. Finally $Z = q^{m-1}(q^m + q - 1) > q - 1 \geq g$, so zero is uniquely maximal. $\square$

All statements are deductive for arbitrary prime powers and nondegenerate bilinear forms. The accompanying prime-field enumeration is only finite counterexample pressure; it is neither experimental nor novelty evidence.

REFERENCES

- [1] Omar Colón-Reyes, Abdul Salam Jarrah, Reinhard Laubenbacher, and Bernd Sturmfels. Monomial dynamical systems over finite fields. *Complex Systems*, 16(4):333–342, 2006. doi: 10.25088/ComplexSystems.16.4.333.
- [2] Sergei V. Konyagin, Florian Luca, Bernard Mans, Luke Mathieson, Min Sha, and Igor E. Shparlinski. Functional graphs of polynomials over finite fields. *Journal of Combinatorial Theory, Series B*, 116: 87–122, 2016. doi: 10.1016/j.jctb.2015.07.003.
- [3] Claudio Qureshi and Lucas Reis. On the functional graph of the power map over finite groups. *Discrete Mathematics*, 346(7):113393, 2023. doi: 10.1016/j.disc.2023.113393.